00 - Steady Master Architecture

Permanent architecture and downstream contracts for Handoffs A through E.

ARCHITECTURAL RULE

Design from Handoff E backward, implement from Handoff A forward. No phase may make a local implementation choice that blocks a downstream requirement without a written Architecture Decision Record describing the tradeoff, migration path, and affected contracts.

1. Product destination

Steady is designed as one system with three surfaces: Steady Personal, Steady Clinical, and Steady Intelligence. The permanent data spine is a longitudinal behavioral-health graph that connects person, state, signal, assessment, need, risk, intervention, response, clinical action, referral, encounter, outcome, utilization, and cost.

Surface	Primary job	Long-term benchmark
Steady Personal	Continuous AI-supported relationship between encounters	Outperform patient-facing behavioral-health companions on continuity, m
Steady Clinical	Prioritized workflow for clinicians and care managers	Match or exceed enterprise clinical workflow and measurement platforms
Steady Intelligence	Population, navigation, outcome, network, and economic intelligence	Match NeuroFlow-class infrastructure, then differentiate through continuo

2. A-E dependency map

Handoff	Build now	Must preserve for later
A	Permanent spine + Steady Personal	Clinical workflow, enterprise tenancy, FHIR/claims/ADT, population models, navigation, outco
B	Personal intelligence + Steady Clinical	Enterprise identity, population scoring, clinician feedback as future learning labels
C	Enterprise + interoperability + population infrastructure	Closed-loop referrals, model governance, utilization and cost linkage
D	Navigation + population learning	Provider/network performance, causal caution, economic attribution
E	Outcomes + cost + orchestration	Uses all prior provenance, event history, data rights, and model versions

3. Permanent domain model

- Identity: person, patient, user, clinician, care manager, provider, organization, payer, employer, care team.
- State and measurement: daily state, clinical state, symptom, sleep, functioning, engagement, assessment, instrument version, score, severity, trend.
- Care: need, goal, care plan, intervention, intervention version, assignment, completion, response, clinical action, care episode.
- AI: memory, observation, pattern, hypothesis, recommendation, inference, evidence, confidence, model, model version.
- Safety: rule, rule version, safety state, gate, escalation, crisis event, cooldown, re-entry.
- Enterprise: tenant, facility, program, contract population, eligibility, benefit, policy configuration.

- Healthcare: encounter, referral, appointment, provider, facility, diagnosis context, medication context.
- Economics: utilization event, claim, allowed amount, paid amount, attribution, episode cost, outcome.

4. Event-first longitudinal architecture

Mutable current-state views may exist for speed, but the authoritative history must preserve events. Examples include daily_checkin.completed, assessment.scored, intervention.completed, intervention.response_recorded, memory.patient_corrected, pattern.proposed, safety_rule.triggered, clinician.reviewed, referral.closed_loop, encounter.received, claim.received, and outcome.measured.

5. Intelligence truth model

Type	Meaning	Authority
FACT	Explicitly supplied by patient, clinician, or trusted external source	Source-backed
OBSERVATION	Directly measured or recorded	Evidence-backed
PATTERN	Derived relationship across observations	Analytical, revisable
HYPOTHESIS	Possible relationship with insufficient evidence	Never presented as fact
RECOMMENDATION	Candidate next action	Advisory
DECISION	Human or deterministic action	Recorded separately from AI recommendation

6. AI architecture

- All model calls pass through a Steady AI Gateway. No feature calls a model provider directly.
- Gateway owns model routing, structured outputs, tool permissions, prompt/version registry, PHI policy, cost, latency, retries, fallback, evaluation, and provenance.
- AI may personalize, summarize, rank candidates, identify candidate patterns, and draft recommendations.
- AI may not clear crisis states, unlock gated pathways, change permissions, erase history, diagnose, prescribe, or autonomously execute consequential clinical actions.
- The generated sentence is disposable. The structured evidence and provenance are permanent.

7. Data governance zones

Zone	Purpose	Default AI use
Operational	Run the product and care workflows	Permitted only for the specific workflow
Patient memory	Patient-controlled longitudinal context	Scoped by memory permissions
Clinical record	Care-team documentation and measures	Strictly role- and purpose-bound
Analytics	Aggregated operational reporting	Governed, minimum necessary
Research	Approved studies and evaluation	Separate consent/IRB/data-use controls as applicable
Model development	Training/evaluation datasets	Never assumed from ordinary product consent

8. Downstream architecture contract

- Versionable and historically recoverable
- Tenant-aware and permissionable
- Provenance attached to source, actor, model/rule version, and timestamp
- Auditable without relying on generated prose
- Correctable by an authorized human without erasing original history
- Exportable and deletable according to policy and data rights
- Usable in future population analysis only under explicit governance
- Capable of linking to outcomes, utilization, and cost when those data arrive
- Compatible with external ingestion and stable IDs
- Independent of any single foundation-model vendor

9. Nonfunctional requirements

- HIPAA-capable security posture and BAA-ready vendors where required; encryption in transit and at rest; least privilege; tenant isolation; immutable security/audit logs.
- Observability IDs across user action, domain command, event, AI call, safety decision, integration transaction, and background job.
- Idempotent ingestion and command processing; retries must not duplicate clinical or economic events.
- Schema migrations are forward-compatible and reversible where practical; data migrations are separately tested.
- Feature flags and model flags allow controlled pilots by tenant/cohort.
- Disaster recovery, backups, retention, deletion, legal hold, and incident response are designed before enterprise pilots.

10. Architecture decision discipline

Any change to canonical IDs, event semantics, tenant boundaries, provenance, AI authority, consent/data rights, intervention versioning, integration contracts, or outcome/cost linkage requires an Architecture Decision Record. ADRs must state the downstream effect on B-E before approval.

A - Foundation + Steady Personal

Build-ready foundation designed to survive every later phase.

ARCHITECTURAL RULE

Design from Handoff E backward, implement from Handoff A forward. No phase may make a local implementation choice that blocks a downstream requirement without a written Architecture Decision Record describing the tradeoff, migration path, and affected contracts.

A0. Objective and exit gate

Build the permanent spine and a best-in-class Steady Personal experience without creating a disposable MVP architecture. Exit when a production-grade companion can operate with deterministic safety, structured memory, measurement, intervention response capture, full provenance, and reserved enterprise/integration contracts.

A1. Repository and service boundaries

- Mobile/web clients consume versioned APIs, never databases directly.
- Domain modules: identity, longitudinal state, measurement, intervention, memory, safety, AI gateway, notifications, consent, audit.
- Shared infrastructure: PostgreSQL, object storage, queue/job system, secrets, observability, feature flags.
- Use a modular monolith initially unless scale evidence requires service extraction. Preserve module contracts so later extraction does not alter domain semantics.

A2. Canonical identifiers and tenancy

- Use immutable UUID/ULID-style IDs for every durable entity; external IDs live in mapping tables.
- Person identity is distinct from account identity and organization enrollment.
- Tenant/organization fields exist now even for direct-to-consumer users; consumer tenancy can be represented by a platform tenant.
- Every future enterprise-owned record has organization scope and explicit sharing semantics.

A3. Core tables / aggregates

Aggregate	Required now	Reserved fields/relations
Person	identity, demographics-minimum, timezone, locale	enterprise enrollments, external identifiers
LongitudinalEvent	event type, actor, occurred_at, recorded_at, payload version	source system, encounter/claim linkage
Assessment	instrument/version, responses, score, severity	population benchmarks, payer program
Intervention	type/version/protocol, contraindications, target state	clinical approval, enterprise formulary
Memory	type, content structure, provenance, confidence, visibility, editability	expiration, organization scope, research/model-use rights
SafetyEvent	rule/version, input evidence, state transition	tenant policy overlay

Aggregate	Required now	Reserved fields/relations
AIInference	task, model/version, prompt/version, structured output, evidence refs	evaluation score, clinician feedback
Consent	purpose, scope, effective/withdrawn dates	research/model development permissions

A4. Memory architecture

- Memory is typed, not raw chat history. Minimum types: goal, preference, trigger, calming resource, coping response, routine, relationship/context, communication preference, clinician-directed context.
- Each memory stores source evidence, creator, timestamp, confidence, sensitivity, patient visibility, patient editability, AI-use permission, scope, and supersession history.
- Patient correction creates a new version and preserves the prior record for audit. The corrected value becomes active.
- Memory retrieval must be purpose-limited. The AI Gateway requests only memory classes required for the current task.

A5. Intervention ontology

- Intervention and InterventionVersion are separate. Never overwrite a protocol used in historical sessions.
- Store modality, target state, expected duration, prerequisites, contraindications, instructions, response measure, clinical approval status, and content provenance.
- Assignments and completions reference the exact intervention version.
- Response capture supports immediate and delayed response so Phase B can learn intervention-response relationships.

A6. Safety state machine

- Deterministic rules execute before generative response where safety-relevant input is present.
- Safety transitions are explicit states, not booleans. Examples: normal, elevated-review, cooldown, blocked-pathway, crisis-routing, re-entry-pending.
- Every rule firing records rule version, evidence, resulting state, user-visible action, and escalation record.
- Generative output cannot override a safety state. Re-entry follows typed criteria and authorization.

A7. AI Gateway contract

Capability	Contract
Structured output	JSON-schema validated; invalid output is rejected/retried/fallback
Tool use	Allowlisted typed tools with explicit risk tier
Context	Purpose-specific retrieval; no unrestricted database context
Logging	Model/version, prompt/version, latency, token/cost, evidence IDs, output hash
Evaluation	Golden sets, safety tests, regression suites, hallucination/grounding checks
Fallback	Deterministic safe response or human escalation when model unavailable/invalid

A8. User experience scope

- Onboarding and consent; daily check-in; text and voice companion; goals; structured memory view/edit/delete; validated measurements; intervention library; journaling/reflection; progress; notifications; safety/crisis routing; history and privacy controls.
- Every AI-generated insight that persists must be represented as a typed object with evidence, not only displayed prose.
- Voice transcripts follow the same provenance, consent, retention, and deletion rules as text.

A9. Reserved interfaces for C-E

- ClaimsAdapter, EHRAdapter, ADTAdapter, ProviderDirectoryAdapter, EligibilityAdapter, ReferralAdapter, SchedulingAdapter, OutcomeAdapter, CostAdapter.
- PopulationModel, RiskModel, RecommendationEngine, ProviderPerformanceModel, OrchestrationEngine.
- Define interface contracts and test doubles now. Do not implement external integrations in A.

A10. Testing and acceptance

- Unit tests for domain rules and state machines; contract tests for AI structured outputs; golden safety cases; property tests for idempotency; migration tests; permission tests; audit/provenance tests.
- Acceptance: a patient can complete six months of simulated activity, correct memory, change consent, trigger and exit safety states, switch model vendors, and retain a coherent longitudinal history without schema redesign.
- Acceptance: replaying events can reconstruct key longitudinal views. Duplicate API retries do not duplicate clinical events.

B - Personal Intelligence + Steady Clinical

Turn longitudinal history into supervised individual and care-team intelligence.

ARCHITECTURAL RULE

Design from Handoff E backward, implement from Handoff A forward. No phase may make a local implementation choice that blocks a downstream requirement without a written Architecture Decision Record describing the tradeoff, migration path, and affected contracts.

B0. Objective and exit gate

Turn history into individual intelligence and make that intelligence useful to a supervised care team. Exit when Steady can show why a patient is prioritized, what evidence supports a pattern, how interventions performed, and how clinician feedback becomes future learning data.

B1. Personal Behavioral Model

- Maintain separate stores/views for facts, observations, patterns, hypotheses, recommendations, and decisions.
- Pattern jobs consume versioned longitudinal events and emit evidence-linked Pattern objects. Patterns are revisable and can be contradicted.
- No diagnosis inference is promoted into the clinical record without authorized human workflow.

B2. Intervention-response engine

- Create pre-state, intervention exposure, immediate response, delayed response, and subsequent-state windows.
- Capture non-response and abandonment, not only successful completion.
- Rank candidate interventions using current state, goals, contraindications, prior response, preference, timing, and clinician constraints.
- Record ranking inputs and chosen action so later evaluation can distinguish recommendation quality from user choice.

B3. Steady Clinical

- Caseload priority screen; patient timeline; measures/trends; alerts; AI summaries; care plans; assigned activities; messages; escalation queue; handoff status; clinical notes.
- Primary workflow answers: who needs attention, why, what changed, what has already been tried, and what happened afterward.
- AI summaries cite underlying events/measurements and never become the sole clinical record.

B4. Clinician feedback labels

Feedback	Future use
Useful / not useful	Recommendation evaluation
Contacted / no contact	Workflow effectiveness
Escalated / no escalation	Routing evaluation
Changed care plan	Decision linkage
Patient improved / worsened / unchanged	Outcome label
Incorrect explanation	Model error taxonomy

B5. Evaluation framework

- Version every model, prompt, intervention, feature, and scoring policy.
- Support cohort assignment, exposure records, outcome windows, and experiment exclusions.
- Predefine product metrics: activation, retention, intervention completion, patient-rated usefulness, clinician-rated usefulness, time-to-review, alert precision, safety false-positive/false-negative review.
- Do not use engagement alone as proof of clinical benefit.

B6. Downstream contract for C-E

- Versionable and historically recoverable
- Tenant-aware and permissionable
- Provenance attached to source, actor, model/rule version, and timestamp
- Auditable without relying on generated prose
- Correctable by an authorized human without erasing original history
- Exportable and deletable according to policy and data rights
- Usable in future population analysis only under explicit governance
- Capable of linking to outcomes, utilization, and cost when those data arrive
- Compatible with external ingestion and stable IDs
- Independent of any single foundation-model vendor
- Clinician feedback must be queryable as labels for future model evaluation.
- Every prioritization score must expose components so C can apply enterprise policy overlays.
- Care-plan and intervention records must accept external/EHR identifiers later.

C - Enterprise + Population Infrastructure

Multi-tenant interoperability, population ingestion, stratification, and measurement-based care.

ARCHITECTURAL RULE

Design from Handoff E backward, implement from Handoff A forward. No phase may make a local implementation choice that blocks a downstream requirement without a written Architecture Decision Record describing the tradeoff, migration path, and affected contracts.

C0. Objective and exit gate

Become enterprise-capable and ingest populations larger than the active Steady user base. Exit when an organization can securely load a population, connect healthcare data, run measurement-based care, and receive explainable stratification without violating tenant or patient data rights.

C1. Multi-tenant enterprise

- Organization, facility, program, contract population, care team, role, permission, policy configuration, eligibility, benefit.
- Tenant isolation enforced at data-access layer and tested with cross-tenant attack cases.
- Consumer records can later enroll into an enterprise program without identity duplication.

C2. Interoperability

- FHIR/SMART on FHIR where appropriate; HL7/ADT where required; claims ingestion; SSO; SCIM; APIs; webhooks; bulk files.
- ExternalIdentifier maps source-system IDs to canonical Steady IDs. Never use external IDs as primary keys.
- Every ingestion transaction is idempotent and records source, file/message ID, schema version, processing status, and errors.

C3. Population identification and stratification

- Support people who do not yet have a Steady account.
- Independent dimensions: clinical severity, deterministic safety state, engagement risk, utilization risk, care-gap risk, predicted deterioration, navigation need.
- Each score stores model/policy version, evidence, confidence/calibration where relevant, and reason codes.
- Enterprise policy can determine thresholds and workflow, but cannot silently change underlying historical scores.

C4. Measurement-based care

- Instrument assignment, cadence, reminders, completion, scoring, severity, change, reliable improvement/worsening, remission where clinically defined.
- Instrument versions are immutable; scoring algorithms are versioned.
- Population reporting separates missingness from improvement.

C5. Enterprise analytics

- Population size and eligibility; activation; engagement; measurement completion; severity distribution; care gaps; escalations; clinical workload; outcomes when available.
- Metrics are defined in a semantic layer so dashboard definitions do not drift between customers.
- Every metric documents numerator, denominator, exclusions, time window, source systems, and freshness.

C6. Downstream contract for D-E

- Versionable and historically recoverable
- Tenant-aware and permissionable
- Provenance attached to source, actor, model/rule version, and timestamp
- Auditable without relying on generated prose
- Correctable by an authorized human without erasing original history
- Exportable and deletable according to policy and data rights
- Usable in future population analysis only under explicit governance
- Capable of linking to outcomes, utilization, and cost when those data arrive
- Compatible with external ingestion and stable IDs
- Independent of any single foundation-model vendor
- Provider and referral entities must support network, benefit, availability, and closed-loop status.
- Claims/utilization events must preserve raw source plus normalized representation.
- Population score history must remain available for retrospective outcome and cost analysis.

D - Navigation + Population Learning

Closed-loop care navigation and governed learning infrastructure.

ARCHITECTURAL RULE

Design from Handoff E backward, implement from Handoff A forward. No phase may make a local implementation choice that blocks a downstream requirement without a written Architecture Decision Record describing the tradeoff, migration path, and affected contracts.

D0. Objective and exit gate

Close the care loop and create governed population learning. Exit when Steady can identify need, navigate to an appropriate provider, confirm what happened, and evaluate population models against real outcomes.

D1. Provider graph

- Provider, organization, specialty, license, location, modality, language, insurance/network, availability, clinical capability, referral rules, quality/outcome metrics.
- Provider identity and network participation are time-bounded. Do not overwrite historical network status.

D2. Closed-loop referral

- Need identified -> care level selected -> candidate providers -> referral -> patient contact -> scheduling -> attendance -> completion/failure -> follow-up -> outcome.
- Failure reasons are first-class data: no capacity, coverage, distance, language, no response, declined, clinical mismatch, scheduling failure.
- Navigation performance becomes measurable.

D3. Navigation AI

- AI may rank provider candidates, explain benefits, draft patient communications, identify barriers, and prepare scheduling handoffs.
- Hard constraints such as licensure, network status, geography, eligibility, and safety policy are deterministic filters.
- Record recommendation set, selected provider, reason, and eventual referral outcome.

D4. Population learning platform

- Separate operational PHI stores from approved analytical/research/model-development datasets.
- Dataset manifests record purpose, consent/legal basis, included fields, cohort definition, extraction date, transformations, and retention.
- Models require training/validation cohort definitions, performance, calibration, subgroup analysis, drift monitoring, deployment approval, and rollback.

D5. Learning Ledger

Question	Required record
What did Steady think?	Prediction/recommendation + model/version
Why?	Evidence IDs + reason codes
What happened?	Outcome window + events
Was it right?	Evaluation result
Did a human agree?	Clinician feedback
Did action change?	Decision and workflow events

D6. Downstream contract for E

- Versionable and historically recoverable
- Tenant-aware and permissionable
- Provenance attached to source, actor, model/rule version, and timestamp
- Auditable without relying on generated prose
- Correctable by an authorized human without erasing original history
- Exportable and deletable according to policy and data rights
- Usable in future population analysis only under explicit governance
- Capable of linking to outcomes, utilization, and cost when those data arrive
- Compatible with external ingestion and stable IDs
- Independent of any single foundation-model vendor
- Referral and provider outcomes must link to utilization/cost episodes later.
- Model predictions must be reconstructable at the point in time they were made, without future-data leakage.
- Analytical pipelines must support causal caution: correlation is not represented as intervention effect without appropriate study design.

E - Outcomes + Cost + Orchestration

Outcome, utilization, network, economic intelligence, and governed next-action orchestration.

ARCHITECTURAL RULE

Design from Handoff E backward, implement from Handoff A forward. No phase may make a local implementation choice that blocks a downstream requirement without a written Architecture Decision Record describing the tradeoff, migration path, and affected contracts.

E0. Objective and exit gate

Connect behavioral state, care actions, outcomes, utilization, and cost into an accountable intelligence layer. Exit when Steady can evaluate care pathways and propose next actions with transparent evidence, while keeping clinical authority and economic incentives governed.

E1. Outcome graph

- Link person -> state -> need -> intervention -> provider -> clinical action -> outcome.
- Outcome definitions are versioned and include source, window, baseline, missingness, and attribution rules.
- Patient-reported, clinician-reported, utilization, and economic outcomes remain distinguishable.

E2. Utilization and cost graph

- Normalize PCP, therapy, psychiatry, urgent care, ED, inpatient, readmission, and other relevant utilization events.
- Claims preserve billed, allowed, paid, member liability, service dates, diagnosis/procedure context, payer, and source.
- Support PMPM, episode cost, behavioral cost, medical cost, and total cost with explicit attribution rules.

E3. Provider and network intelligence

- Evaluate access, referral completion, wait time, outcome, utilization, and cost by provider/network/cohort.
- Never collapse provider performance into a single opaque score without component visibility and risk adjustment.
- Historical provider/network state is preserved for fair retrospective analysis.

E4. Care-pathway intelligence

- Represent sequences of interventions, referrals, clinical actions, and utilization as analyzable pathways.
- Compare outcomes across pathways with cohort adjustment and clear limits on causal interpretation.
- Identify bottlenecks, failed referrals, unnecessary escalation, under-treatment, and potentially lower-intensity effective pathways for further study.

E5. Economic intelligence

- Cost per active patient, cost per improved patient, cost per remission where defined, PMPM change, acute utilization, clinical labor per patient, outcome by pathway, cost by pathway.
- Savings claims require predefined comparator, observation window, risk adjustment, statistical method, and independent review appropriate to the claim.

E6. Orchestration engine

- Inputs: personal intelligence, clinical state, deterministic safety, population models, care availability, benefits, outcomes, cost, enterprise clinical policy.
- Outputs are candidate next actions such as continue support, increase monitoring, surface to care manager, assess, refer, request review, increase/decrease intensity.
- Every action has an authority tier: informational, patient-choice, clinician-approval, or prohibited autonomous action.
- Economic optimization may never override clinical safety or access policy.

E7. Final system acceptance

- A complete patient journey can be reconstructed from original evidence through AI recommendations, human decisions, referral, outcome, utilization, and cost.
- Models can be replaced without losing longitudinal meaning.
- A patient can exercise applicable data rights without corrupting enterprise audit obligations.
- A payer/health system can understand why Steady prioritized a person or pathway.
- Steady can state what it knows, what it infers, what remains uncertain, and whether prior predictions were correct.